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Assignment 22

Naive Bayes

Email Spam Detection

S1 See Dataset

S2 Train Naive Bayes

A) Prior Probabilities

P(Spam) = 3/6 = 0.5

P(Not Spam) = 3/6 = 0.5

B) Likelihood probabilities

For spam

P(Offer = Yes | spam) = 2/3

P(Money = Yes | spam) = 2/3

P(Meeting = Yes | spam) = 0/3

Not Spam

P(Offer = Yes | Not spam) = 1/3

P(Money = Yes | Not spam) = 0/3

P(Meeting = Yes | Not spam) = 3/3

Step3 : New Email Prediction

New Email

Offer = Yes

Money = Yes

Meeting = No

S4 Apply Naive Bayes
for spam
$P(spam|x) \propto (\frac{2}{3} \times \frac{2}{3} \times 1) = 0.5$

$P(Not\ spam|x) \propto (\frac{1}{3} \times 0 \times 0) = 0$

Predict spam
Predict for all mails

Email	Actual	Predicted	
1	spam	spam	✓
2	spam	spam	✓
3	spam	Not spam	✗
4	Not spam	Not spam	✓
5	Not spam	Not spam	✓
6	Not spam	spam	✗

S5 Confusion Matrix

	Predicted spam	Predict Not spam
Actual spam	$T_P = 2$	$T_N = 1$
Actual Not spam	$F_P = 1$	$F_N = 2$

S6 Understanding each value
T_P (True Positive) = 2 spam identified ✓
T_N (True Negative) = 2 Not spam identified correctly
F_P (False Positive) = 1 Not spam wrongly marked spam
F_N (False Negative) = 1 Spam missed

S7 Calculate Metrics

$$Accuracy = \frac{T_P + T_N}{T_P + T_N + F_P + F_N}$$

$$\text{Recall} = \frac{TP}{TP + FN} = \frac{2}{3} = 66.7\%$$

Conclusion

FP is dangerous → Important Mail marked spam

FN is dangerous → Spam Reaches Inbox

So depending
Email system shd reduce FP & FN